

Reverse engineering

Introduction

Ruxandra F. Olimid

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Agenda

- Course Organization
- Grading
 - Passing Criteria
- Topics

Course Organization/Grading

- Course
 - 2h every 2 weeks
- Laboratory
 - 2h every week
 - **Take care!** Lab work min.45% is required to pass the exam!
 - **Take care!** You cannot redo the lab work at the exam!
- **Course Syllabus:** <https://cursuri.fmi.unibuc.ro/>
- **Moodle:** [Reverse engineering and exploitation \(2025-2026\)](#)
 - All info, materials, students, grades, etc.
 - **Passing criteria**

Topics covered

The list of notions you need to know/understand for the exam
(FROM LAST YEAR)

- General notions such as assembly, compile, link, load, binary, machine code, libraries (static vs, dynamic), debugging, offsets, etc.
- Assembly (language, registers, flags, stack, heap, syscalls, etc.)
- ASLR
- Black-box vs white-box (vs.gray-box) analysis
- Canaries (stack canaries)
- Compiler optimization
- ELF (including the structure, general and specific notions such as segments, sections, GOT, PLT, etc.)
- dex files

- Immediate vs. lazy binding
- NX
- PIE
- Physical vs. logical addresses
- RELRO
- ROP
- Symbols
- Static vs. dynamic binaries/linking/analysis

Problems?

- Contact me or the lab assistant in time!

Take away message:

Attend the laboratory and check Moodle!